

Maanvi Bansal

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EDUCATION

Thapar Institute of Engineering & Technology

Patiala, India

Bachelor of Engineering, Electrical Engineering | Minor: Computer Science Engineering

Sep 2022 - Jun 2026

Relevant Courses: Data Structures & Algorithms, Design & Analysis of Algorithms, Machine Learning, Artificial Intelligence, Data Analytics

PUBLICATIONS

J. Ban, R. Pal, **M. Bansal**, and S. Pal, “Beyond Fixed Thresholds: Adaptive Confidence-Gated Pseudo-Labeling and Patch Consistency for Cross-Modality Medical Segmentation,” **Accepted** at the International Conference on Image Processing Theory, Tools and Applications (IPTA), 2026, proceedings forthcoming.

M.Bansal, G.S. Madhuri, S. Muhuri, “When One Answer Breaks, What Survives? Source Conditioned Cross Task Transfer in Remote Sensing VLMs”, **In Review** at International Conference on Computation, Communication & Intelligent Systems in Engineering 2026, Kolkata, India, IEEE Conference ID: 73082 Acceptance Date: 30 November

M.Bansal, G.S. Madhuri, S. Muhuri, “BEN-10: Cross-Country Dissonance” **In Review** at IEEE International Conference on Smart Innovations in Systems, Infrastructure, Mechanical, Power, AI and Computing Technologies (SISIMPACT RECORD NO. 73129) Acceptance data: 5 November

WORK EXPERIENCE

Machine Learning Intern | CNH Industrial

Feb 2026 - Present

- Architecting a multi-source satellite imagery evaluation framework across Sentinel-2, Google Maps, and OSM tiles, quantifying source quality via spatial resolution, cloud-cover distributions, revisit cadence, and ingestion reliability to determine optimal inputs for field boundary detection at scale.
- Developing automated preprocessing pipelines standardizing CRS, resolution, and bounding boxes across geographically diverse agricultural regions, enabling systematic cross-source model benchmarking.
- Prototyping image fusion methods pixel blending, pyramid fusion, and super-resolution evaluating each variant against field boundary model outputs using IoU, Dice score, and boundary precision/recall to identify performance gains over single-source baselines.

Machine Learning Research Summer Intern | CNH Industrial

Jun 2025 - Aug 2025

- Benchmarked zero-shot and supervised segmentation approaches for field-boundary extraction on satellite imagery, comparing SAM, U-Net, and DeepLabV3+ against verified ground truth using IoU, boundary overlap, and distance-based evaluation metrics.
- Conducted model and imagery analysis across 3 key customer accounts (~1,500 farms), identifying data-quality and geographic variability challenges affecting segmentation and predictive modeling robustness.

Product Manager (Student Intern) | Thapar Institute (Faculty & Placement Cell Collaboration)

Oct 2024-Jun 2025

- Led discovery with 50+ stakeholders to map end-to-end hiring workflows into clear product requirements, translating technical findings into executive-ready recommendations for engineering and business teams.
- Worked with an 8-member team to write requirement specs, API contracts, data field schemas, edge-cases, and dashboard requirements, improving clarity for engineering implementation and business objectives.

ONGOING RESEARCH PROJECTS

Can vision-language models read the sundial? A Blender benchmark of time inference from overhead shadows.

Sep 2026 - Present

- Under the mentorship of **Dr. Samya Muhuri**, Thapar Institute of Engineering to design Given two overhead renders of the same static scene at different times of day using Blender

Spatio-Temporal analysis of 4D structure using Blender

Oct 2026 - Present

CERTIFICATIONS

Hyperspectral Data for Land and Coastal Systems	NASA ARSET
Spatial Data Science: The New Frontier in Analytics	Esri Training MOOC
NOAA's TEMPO Aerosol Products for hourly Aerosol Monitoring over North America	NASA ARSET
Flood Mapping with NISAR	NASA ARSET

SKILLS

Languages: C, C++, Python, SQL, R, Assembly, LaTeX

Machine Learning: PyTorch, TensorFlow, NumPy, Matplotlib, Scikit-learn, Pandas, PySpark, CUDA

Tools & Cloud: Linux (Kali, Ubuntu/WSL), Azure DevOps, Blender, BlenderGIS, ARCGIS Pro, QGIS

POSITIONS OF RESPONSIBILITY

Founding Member and Product Lead | PLEX, TIET

2023 - 2025

- Mentored 50+ students through workshops, case studies, and hands-on projects to instill product thinking and strategy.

Student Mentor | Frosh (Admissions Cell), TIET

2023 - 2024

- Guided 3,000+ applicants during Thapar’s offline admission process, guiding them through enrollment and campus onboarding.